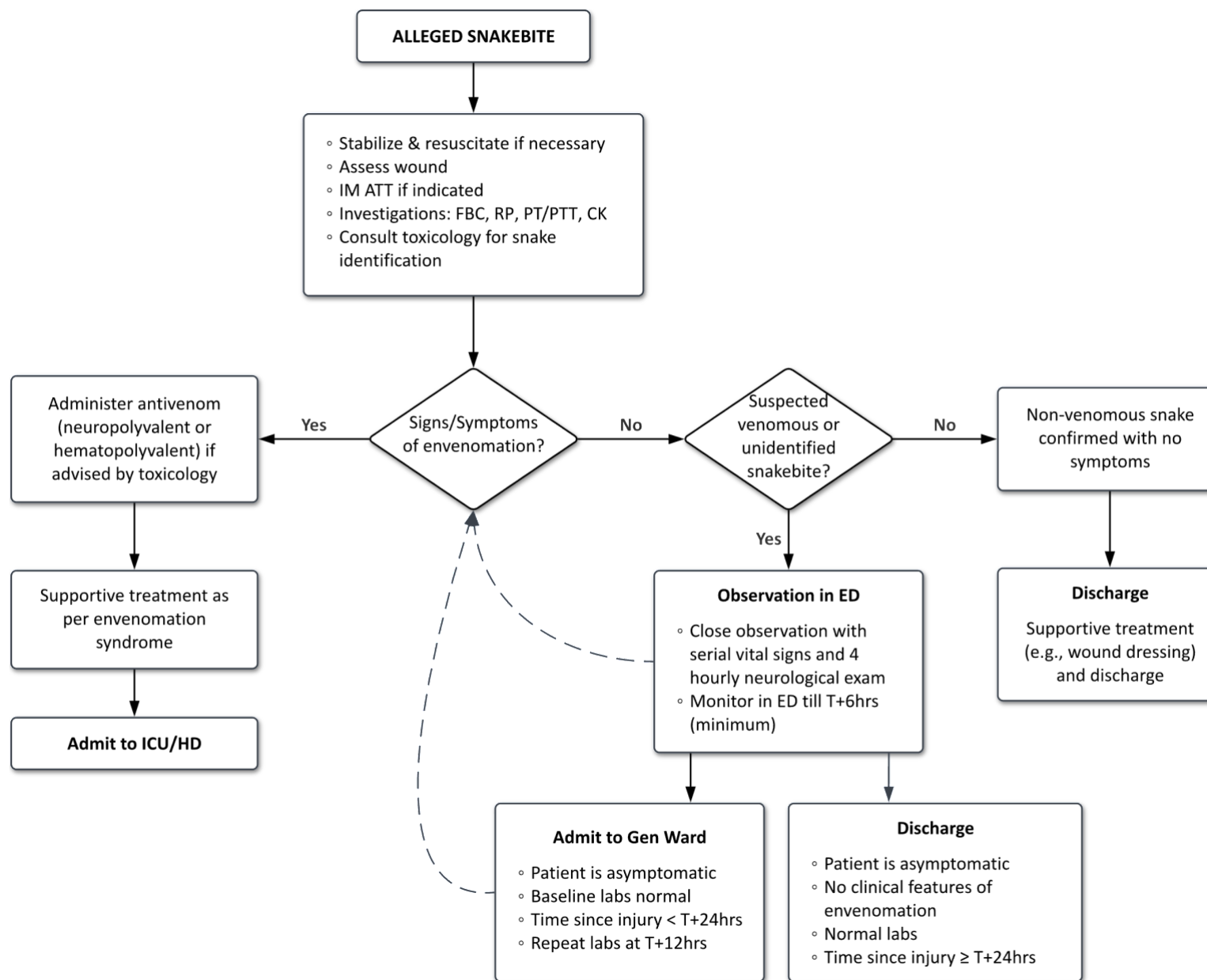


MANAGEMENT OF SNAKEBITES

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MANAGEMENT OF SNAKEBITES

1. BACKGROUND

- The majority of the 78 snake species in Singapore are non-venomous. >25% of bites by venomous snakes are dry bites & do not result in systemic envenomation.
- Attempts at snake identification should be performed if possible. Consult a clinical toxicologist for identification and management advice.
 - In an unknown or suspected snakebite, treatment is determined by clinical presentation and evolution of signs & symptoms

1.1 ENVENOMATION

- Early non-specific features of envenomation may include headache, nausea, vomiting and abdominal pain
- The **3 major clinical syndromes** of snakebite envenomation are:
 - Neurotoxic: bilateral ptosis, cranial nerve deficits, descending paralysis, dyspnoea/AMS
 - Haemotoxic: local bleeding from mucosal sites, coagulopathy, systemic bleeding
 - Cytotoxic: severe pain, progressive edema, tissue destruction, myotoxicity/rhabdomyolysis

2. CLINICAL ASSESSMENT

2.1 HISTORY

- Date, time and location of snakebite
- Details of snakebite: no. of bites, body part affected, description of snake (obtain photos if available to aid identification)
- First aid rendered so far (if pressure bandage immobilization was applied etc.)

2.2 ASSESSING FOR ENVENOMATION

2.2.1 NEUROTOXIC SYNDROME

- Descending flaccid paralysis (Ptosis, ophthalmoplegia, dilated pupils, neck flexor muscle weakness, dysphagia, difficulty speaking)
 - Ptosis is the first sign
- Respiratory weakness
 - Serial capnography & blood gas
 - Single breath count (deep breath → count upwards with normal speaking voice); correlates well with FVC in ≥ 18 years (Normal: 1 to >20, abnormal: Under 15)
 - NIF meter (abnormal if less than -30 or serially declining)

2.2.2 HAEMOTOXIC SYNDROME

- Inspect wound & venipuncture sites for persistent local bleeding
- Examine for mucosal bleeds, haematuria, haematemesis, intracranial haemorrhage
- Decision for antivenom is clinical – do not wait for coagulation panel results

2.2.3 CYTOTOXIC SYNDROME

- Mark leading (proximal) edge of edema with micropore tape → repeat every 1 – 2 hours and document rate of proximal progression (RPP) in cm/hr
 - Ultrasound can be used to accurately assess the leading edge of edema from the cobblestone appearance of soft tissues
- Document pain score progression
- Tender lymphadenopathy of draining lymph nodes
- Local envenomation can cause tissue necrosis and mimics compartment syndrome. Early administration of antivenom is an effective treatment, avoiding the need for fasciotomy which has not been shown to improve outcomes.

Adapted from Table 1: Signs & Symptoms of Envenomation by Venomous Snakes in Malaysia & Singapore (Guide to the Essentials of Emergency Medicine Chapter 96: Snake-related injuries)

Signs	Cobra	Krait	Coral	Pit Viper	Sea Snake	Keelback
Cytotoxic Effect						
Pain, edema, tissue necrosis, blistering	+	–	–	+	–	–
Neurotoxic Effect						
Ptosis, CN deficits, neck and respiratory muscle weakness	++	++	+	–	+	–
Haemotoxic Effect						
Spontaneous bleeding from mucosal sites, hemoptysis, hematemesis, hematuria etc.	–	–	–	+	–	+
Rhabdomyolysis						
Myalgia, myotoxicity and rhabdomyolysis	–	-	+/-	–	++	–

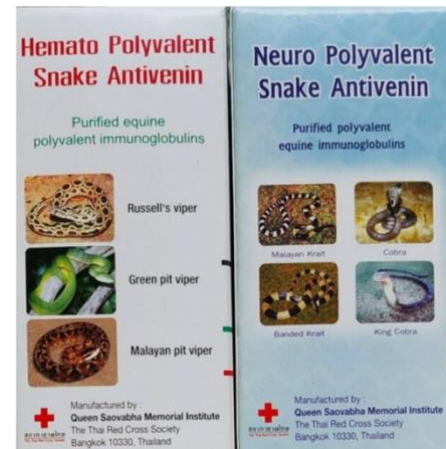
3. INVESTIGATIONS

- Full blood count*
- Coagulation studies (PT/PTT/INR)*
 - If coagulopathy present INR >1.2, check D-dimer and fibrinogen
- Renal panel #1 (ext.)
- Creatine kinase*
- ECG, VBG/CBG, UC10

**Should be repeated at T+6 and T+12 hours after antivenom if administered*

4. ANTIVENOM THERAPY

- There are **NO ABSOLUTE CONTRAINDICATIONS** for antivenom.
- Dosing is not weight based; there is no difference in dosing between adults and children as venom injected is the same.
- There are 2 polyvalent snake antivenoms available in NUH:
 - **Thai Neuro Polyvalent Antivenom (NPAV)** – covers King cobra, monocled cobra, Malayan krait and banded krait
 - **Thai Haemato Polyvalent Antivenom (HPAV)** – covers Russell's viper, green pit viper and Malayan pit viper
- Antivenom should be administered as soon as clinically indicated (best within 6 hours post bite), consult toxicology on call for advice.
- Sea snake antivenom is not stocked in NUH.



4.1 RISK OF ANAPHYLAXIS

- Risk of anaphylaxis from antivenom is <5%
- Pre-medication is not required
- Standby agents for treatment of anaphylaxis prior to administration of antivenom:
 - IM Adrenaline (1:1000) 0.01 mg/kg (max. 0.5mg)
 - IM/IV Diphenhydramine 1 – 2 mg/kg (max. 50mg)

4.2 INDICATIONS FOR ANTIVENOM THERAPY:

- Cytotoxicity
 - Significant local pain (PSP: pain score progression)
 - Progressive edema (RPP: rate of proximal progression) rapidly extending within a few hours of bites on hands or feet
 - Signs of tissue destruction (blistering, necrosis)
- Neurotoxicity
 - Onset, or progression of neurotoxic signs
 - Deteriorating respiratory function
- Haemotoxicity
 - Spontaneous bleeding
 - Venom induced consumptive coagulopathy (INR >1.2)
- Rhabdomyolysis
- Acute Kidney Injury
- Cardiovascular instability, cardiac arrest, arrhythmias

4.3 ADMINISTRATION OF ANTIVENOM

4.3.1 ANTIVENOM RECONSTITUTION

- Reconstitute each vial of freeze-dried antivenom with the solution supplied (or 10ml of sterile water for injection)
- Gently swirl (not shake) well until serum becomes a clear solution
- Reconstituted solution is further diluted with IV N/S and infused over 30mins
 - Child: 5ml/kg body weight
 - Adult: 250ml

ANTIVENOM	INITIAL DOSE ¹
Neuro Polyvalent Antivenom	5 vials (50ml)
Haemato Polyvalent Antivenom	3 vials (30ml)

4.3.2 MONITORING

- Close monitoring during and for at least 1 hour after completion of antivenom
- Re-evaluate at 6- and 12-hours post antivenom administration with repeat bloods
- Cardiac and urine output monitoring for venom induced cardiotoxicity and renal impairment

4.4 MANAGEMENT OF REACTIONS TO ANTIVENOM

TYPE OF REACTION	CLINICAL FEATURES	TREATMENT
Mild or moderate reaction during infusion	Nausea, vomiting, urticaria, chills, fever	Stop the infusion Manage symptomatically with antipyretics, antiemetics, antihistamines etc. Reassess the patient once the reaction has been controlled, if meets criteria for antivenom, resume infusion at a slower rate.
Anaphylaxis	Pruritus with urticarial rash Cough, nausea/vomiting, diarrhea, abdominal colic Hypotension, angioedema, bronchospasm Adrenaline should be administered at the earliest sign of anaphylaxis	Stop the infusion. Manage anaphylaxis. IM adrenaline (1:1000) 0.01ml/kg every 5 – 10 mins IV adrenaline infusion @ 1mcg/kg/min (if deteriorates further) Reassess the patient once the reaction has been controlled, if meets criteria for antivenom, resume infusion at a slower rate.
Pyrogenic reaction [Early] (1 – 2h after treatment)	Chills/rigors, fever, vasodilation with hypotension May precipitate seizures in children	Active cooling with antipyretics. IV fluids for hypovolemic state.
Serum sickness [Delayed] (1 – 12 days [mean 7 days] after treatment)	Fever, nausea, vomiting, diarrhea, itching, recurrent urticaria Arthralgia, myalgia, rash, lymphadenopathy (influenza-like illness)	Regular antihistamines x 5 days. Prednisolone 30-50mg/day (1mg/kg in paediatrics up to 30mg) daily x 5–7 days

¹ Taken from WHO Guidelines For The Management of Snakebites, 2nd Edition (2016), Chapter 6 – Management of Snakebites in South-East Asia, Page 141, "Table 1: Guide to initial dosage of some antivenoms for treating bites by medically-important snakes in the South-East Asia Region, based on clinical experience, rather than manufacturers' recommendations."

5. ADJUNCT TREATMENT & SUPPORTIVE CARE

5.1 WOUND CARE

- Remove restrictive clothing & clothes covering the wound
- Do not apply constrictive dressings or tourniquets that will worsen tissue injury
- Immobilize limb loosely without adding pressure to the soft tissue (to contain local spread)
- If pressure dressing/tourniquet already applied in the pre-hospital setting:
 - Do not remove until team is prepared for resuscitation and potential deterioration
 - Remove tourniquet gradually (loosen → tighten → observe) over 20 minutes
 - If symptoms develop, administer antivenom → wait 30 mins before releasing tourniquet again

5.2 ANTIBIOTICS

- Evidence does not support use of empiric antibiotics;
- Wound culture should be performed only for clinically infected wounds prior to antibiotics for goal-directed antimicrobial therapy. Random wound cultures are not recommended.
- Empiric therapy should cover for enteric gram-negative bacteria (*Salmonella* spp., *Yersinia*, *Campylobacter*, *Aeromonas*), anaerobes and skin organisms

5.3 IMMUNIZATION

- Tetanus toxoid should be given. Check patient's immunization status against tetanus

History of Adsorbed Tetanus Toxoid ²	DTap, Tdap, or Td	TIG
< 3 doses or unknown	Yes	Yes
≥ 3 doses	No, if < 5 years since last tetanus containing vaccine	No
	Yes, if ≥ 5 years since last tetanus containing vaccine dose	No

Tdap: tetanus toxoid, reduced diphtheria toxoid, and acellular pertussis vaccine;

DTaP: diphtheria and tetanus toxoids and acellular pertussis vaccine;

Td: adult-type diphtheria, and tetanus toxoids vaccine;

TIG: Tetanus Immune Globulin (human).

² Table adapted from Red Book 2018 – 2021 Report of the Committee on Infectious Diseases, 31st Edition, Table 3.78 Guide to Tetanus Prophylaxis in Routine Wound Management, Page 796.

5.4 NEUROTOXIN SYNDROME

- Intubate and ventilate if significant muscle weakness with impaired ventilation
- Anticholinesterase drugs may be useful in neurotoxic envenomation
 - Atropine 0.6mg (adults) or 0.01mg/kg (paeds) followed by Neostigmine 2.5mg (adults) or 0.05mg/kg (paeds)
 - Combination mixture (for paeds) of:

Drug (Concentration)	Volume (mL)	Dose (mg)	Remarks
Neostigmine (2.5 mg/mL)	0.3	0.75	0.1ml/kg of solution contains: - Neostigmine <u>0.05 mg/kg</u> - Atropine <u>0.01 mg/kg</u>
Atropine (0.6 mg/mL)	0.25	0.15	
Normal Saline (0.9% NaCl)	0.95	-	
Total volume	1.5	-	

- Do not repeat if no effect after 1st dose
- If reversal of paralysis observed, repeat every 2H to 4H (max. dose of Neostigmine = 5mg per 24h)

5.5 HAEMOTOXIN SYNDROME

- Transfuse blood products only in life-threatening bleeding
- Tranexamic acid and Vitamin K are not useful in venom-induced consumptive coagulopathy
- Normalization of coagulation profile lags behind clinical response by 24-48 hrs

5.6 VENOM OPHTHALMIA (NAJA SUMATRANA – BLACK SPITTING COBRA)

- Symptoms of intense pain with risk of vision loss
- Principles of management:
 - Decontaminate: Copious amounts of normal saline
 - Analgesia: Topical tetracaine 0.5%
 - Antibiotic prophylaxis: Topical antibiotics for prevention of endophthalmitis
 - Complications: Refer EYE for evaluation of cornea, intra-ocular pressure
- Topical use of antivenom is not indicated for venom ophthalmia

6. DISPOSITION

6.1 ADMISSION

- Snake bites with local or systemic envenomation requiring antivenom (admit to HD/ICU)
- Bite by a venomous snake OR unidentified snakebite without signs of envenomation at 6 hours (admit to general ward for observation)

6.2 DISCHARGE

- Bite by a non-venomous snake that has been identified by a trained expert, asymptomatic and no significant tissue injury such as lacerations or persistent bleeding
- Bite by a venomous snake OR unidentified snakebite and has remained asymptomatic for 24 hours after injury with no clinical evidence of envenomation

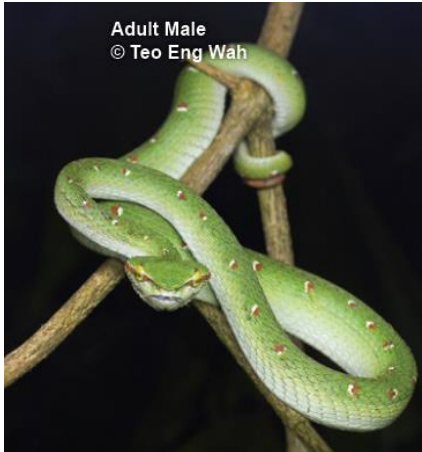
ANNEX A: OVERVIEW OF TREATMENT OPTIONS FOR ENVENOMATION

SYSTEM	CLINICAL FEATURES	TESTS	TREATMENT	MONITORING
Local Tissue	Pain Blistering, tissue necrosis, ecchymosis Tender regional LN Rapidly ascending edema		Antivenom (only if limb threatening or rapidly progressing) Compartment syndrome is rare.	Pain score progression Rate of proximal progression (edema)
Cardiotoxicity	Hypotension Tachycardia Shock Cardiac arrest	12-lead ECG Troponin I	Antivenom IV fluids (isotonic solution or blood) Support with inotropes if necessary	HR, BP
Neurotoxicity	<u>Ptosis</u> Diplopia Dysphagia Bulbar palsy Neck flexor muscle weakness Dyspnea	ABG – T2RF	Antivenom Atropine & neostigmine (for descending flaccid paralysis) - Can be given q2 – 4H Secure airway & support if necessary *Paralysis reversed with anticholinesterase is likely to respond to antivenom.	RR, SpO2 Capnography Serial blood gases Older patient – Single Breath Count (SBC)
Coagulopathy	Epistaxis Oozing from wound site, venipuncture site Ecchymosis	FBC PT/PTT/INR D-dimer, fibrinogen GXM	Antivenom (primary tx) Blood products only if life-threatening bleed – otherwise hold till after antivenom.	Normalization of coagulation profile in 6H (lag behind clinical response)
Rhabdomyolysis Nephrotoxicity	Dark urine Myalgia	UC10 / UFEME myoglobin CK/AST/ALT RP2 ECG	Antivenom Intravenous hydration, maintain UO > 4 ml/kg/h. *Antivenom does not reverse the rhabdomyolysis process but attenuates it	Serial CK, RP

Warrell DA. Envenoming and injuries by venomous and nonvenomous reptiles worldwide. In: Wilderness Medicine, 6th Edition, Auerbach PS. (Ed), Elsevier Mosby, Philadelphia 2012. p.1040

ANNEX B: MEDICALLY SIGNIFICANT SNAKES IN SINGAPORE

- There are 7 venomous land snakes and 1 venomous sea snake in Singapore.
- 2 cobras, 2 kraits, 2 coral snakes and 2 vipers.

Family	Species	
Viperidae	Wagler's Pit Viper (<i>Tropidolaemus wagleri</i>)	
		
	Mangrove / Shore Pit Viper (<i>Trimeresurus purpureomaculatus</i>)	
		

Family	Species	
Elapids	Sunda King Cobra (<i>Ophiophagus bungarus</i>)	Black Spitting Cobra (<i>Naja sumatrana</i>)
		
	Banded Krait (<i>Bungarus Fasciatus</i>)	Yellow Lipped Sea Krait (<i>Laticauda Colubrina</i>) Sea snake
		
	Malayan Blue coral snake (<i>Calliophis bivirgatus</i>)	Malayan Banded coral snake (<i>Calliophis intestinalis</i>)
		